

Mexicali Urban Liveability Index

How the index is calculated

Generated 2026-09-25 · Companion to the liveability dashboard.

The index is provisional: the indicators in each domain and how it is calculated are still under review.

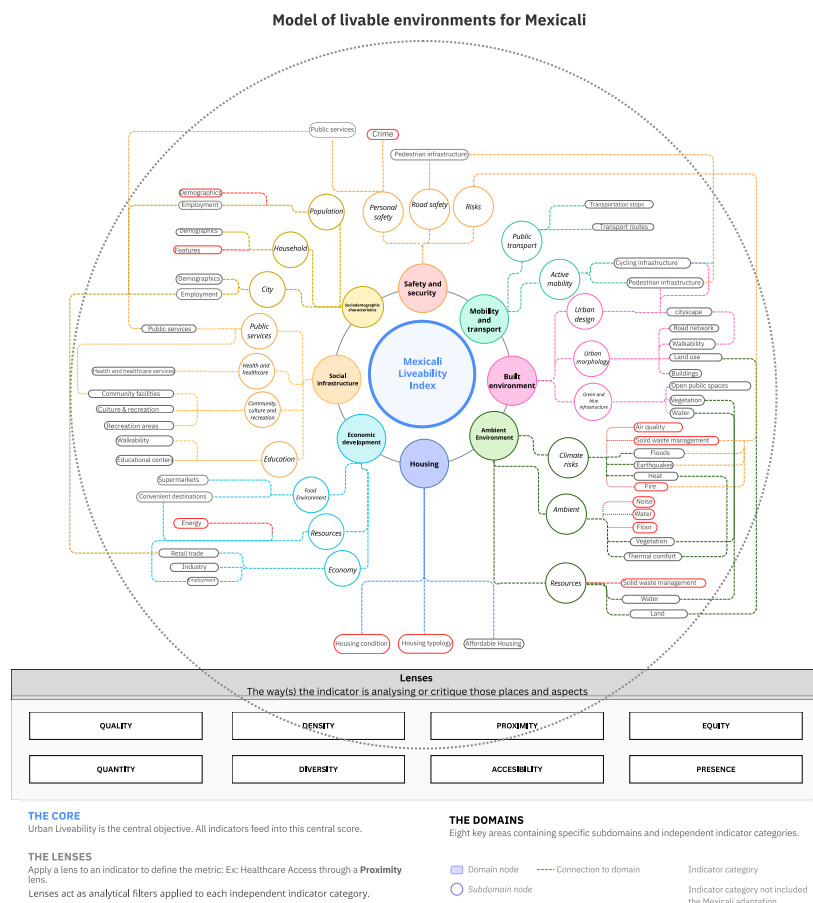
Overview

Average of the Mexicali Urban Liveability Index's domain scores, each summarising its normalised sub-indicators, less a penalty for variation across domains that rewards balanced performance. Expressed as points above or below the study region average (0).

The Urban Liveability Index draws on the Adjusted Mazziotta-Pareto Index (AMPI): each sub-indicator is re-scaled so that 0 is the study region average (its mean over all sample points), and scores are points above or below it. Each domain is the weighted mean of its sub-indicators less a penalty for imbalance between them, and the index is likewise the mean of the domains less a penalty for imbalance between domains. A sub-indicator bearing on several domains counts an equal share of its weight in each (a third in each of three), so that in all it weighs no more than one in a single domain. Scored at sample points along the pedestrian network, and averaged to grid cells and, weighted by population, to larger areas.

Mexicali Urban Liveability Index summarises 28 indicators estimated for every sample point on the pedestrian network, grouped by the domains of the conceptual model: Ambient environment, Built environment, Economic development, Mobility and transport, Safety and protection, Social infrastructure. It follows the approach of the Urban Liveability Index (Higgs et al. 2019): a Mazziotta-Pareto composite index, which is partially non-compensatory, so that doing well on some aspects cannot wholly make up for doing badly on others.

Conceptual model



Domains

● **Ambient environment** · 9 indicators

The quality of environmental conditions to which residents are exposed every day. Examples include air, water, noise and temperature.

● **Built environment** · 8 indicators

The physical environment created by people, whose density, mix of land uses and connectivity shape everyday life. Examples include buildings, streets and public spaces.

● **Economic development** · 6 indicators

Local economic conditions that offer residents opportunities for productive employment, fair income and economic participation.

● **Housing** · no indicators yet

The availability, affordability and suitability of housing options that meet the diverse needs of neighbourhood residents.

● **Mobility and transport** · 10 indicators

Networks and systems that enable efficient travel between home, work and services.

● **Safety and protection** · 5 indicators

Conditions in the urban environment that allow residents to move freely and participate in public life without fear of crime, violence or traffic accidents.

● **Social infrastructure** · 13 indicators

The network of public facilities that supports social interaction, cohesion and collective well-being.

● **Sociodemographic characteristics** · no indicators yet

Characteristics of the population, such as age, income, educational attainment and household composition, that determine how quality of life is perceived across different groups.

1. From measurement to indicator

The distance d to the nearest destination is first turned into a soft access score between 0 and 1, using the soft threshold of Higgs et al. (2019): a logistic function of the threshold t (300 m, 1.5 km, 2 km, 3 km) with slope $k = 5$. It is 0.5 at the threshold, close to 1 well within it (0.99 at $d = 0$) and close to 0 well beyond it (0.007 at $d = 2t$), and 0 where nothing was found within the distance searched. Unlike a binary cut-off, a small difference in distance near the threshold makes only a small difference to the score.

$$s(d) = 1 / (1 + e^{k(d-t)/t})$$

Some distances are scored instead by a published ladder: Natural blue space nearby (100 within 200 m, 90 within 500 m, 80 within 1 km, 60 within 1.5 km; 60 beyond). Indicators that are already proportions, diversity scores or indices enter as they are.

2. Normalisation

Each indicator is re-scaled using the normalisation of the Adjusted Mazziotta-Pareto Index (AMPI; Mazziotta and Pareto 2018), reversed where lower values are better so that a higher score always means more liveable, and expressed as points above or below the reference. The reference Ref is the indicator's mean across all the study region's sample points, and scores 0. The goalposts Min and Max are centred on Ref and as far apart as the indicator's observed range, so that range spans 60 points (–30 to +30 when the mean lies midway). An indicator with a scale of its own, such as an index from 0 to 100, may use that scale as its goalposts instead, in which case its observed values span fewer than 60 points. The goalposts used are recorded: scoring a later analysis against the same goalposts makes scores comparable over time.

$$r = 100 \pm 60 \cdot (x - \text{Ref}) / (\text{Max} - \text{Min}), \text{ by default, } \text{Max} - \text{Min} = \max(x) - \min(x) \text{ } (\pm : + \text{ where higher is better, } - \text{ where lower is better})$$

Goalposts resolved on 2026-09-25 from 153,309 sample points.

3. Aggregation: domains and index

Within each domain, the weighted mean M and standard deviation S of its indicators' scores are found (with weights w that sum to 1: equal, unless configured otherwise or an indicator is shared with other domains, when it counts only its share). The score is $M - S \cdot cv$, where $cv = S/M$ is their coefficient of variation: the penalty $S \cdot cv$ grows with the imbalance between indicators, so that a balanced profile scores higher than an uneven one with the same mean (De Muro, Mazziotta and Pareto 2011; weighted form, Mazziotta and Pareto 2022). The index is calculated the same way from the domain scores, without normalising them again, so each domain carries the same weight whatever its number of indicators. As cv divides by the mean, these are calculated with the reference at 100, where every score is positive, and the result is then expressed relative to the reference (less 100).

$$M = \sum w_i r_i, \quad S = \sqrt{(\sum w_i (r_i - M)^2)}, \quad cv = S / M$$

$$w_i \propto \text{weight}_i \cdot \text{share}_{i,d}, \quad \sum w_i = 1 \quad (\text{share} = 1/k \text{ for an indicator in } k \text{ domains})$$

$$\text{score} = M - S \cdot cv \rightarrow \text{reported as score} - 100 \text{ (0 = study region average)}$$

Some indicators bear on more than one domain of the conceptual model. Such an indicator counts a share of its weight in each domain it belongs to (a third in each of three), so that its shares add up to the weight of any other indicator; its score is the same in each. Because every domain carries the same weight in the index whatever it

holds, an indicator's effective weight in the index still depends on how many others share its domains: here from 1.7% to 6.2% of the index's mean level. Indicators shared between domains also make their scores move together, which reduces the penalty for imbalance between domains (though not within them).

4. From sample points to areas

The index is calculated at each sample point and then averaged: to each grid cell as the mean of its points, and to larger areas and the region as averages, usually weighted by population. An area's index is therefore the average of its points' indices, not the index of its average values, and its penalty is the average imbalance at its points.

This dashboard shows values as points above or below the reference (0): a place where every indicator equals its average over the study region's sample points. Because the penalty is subtracted everywhere, and areas are weighted by population, the region's own index is not exactly 0. Each indicator is seen through a lens (proximity, accessibility, density...).

5. Walkability and thermal comfort

Walkability sums the z-scores of access to daily living (a fresh food market, a convenience store and public transport, within 300 m), and population density and intersection density within a 1 km walk (Frank et al. 2010), taken across all sample points. By default it is attenuated by daytime thermal comfort: its percentile rank is multiplied by $1 - \lambda \cdot h$, where h is mean daytime UTCI re-scaled from 0 at 30 °C to 1 at 50 °C, and $\lambda = 0.5$. Heat reduces how walkable a place is, so shade cannot make up for having nothing to walk to, after measures of accessibility in which heat stress shortens the distance people walk.

Walkability can be shown without thermal comfort attenuation. Thermal comfort then enters instead as an ambient environment indicator, so that it counts once and only once: the index is calculated each way, each against its own goalposts.

Daytime thermal comfort (UTCI) was modelled for the hottest day of 2023 over pedestrian surfaces. Air temperature, humidity and wind are uniform across the city, so only radiation (sun and shade) varies; most values exceed the range the UTCI polynomial was fitted on. Compare nearby places with each other, not absolute values or distant neighbourhoods.

The index rewards a balanced profile by design: every domain carries the same weight. The settings allow other weights, to explore what matters to each person; this expresses personal preference, not public health evidence. With custom weights, scores are recalculated in the browser from each area's average scores, so they are an exploratory approximation and do not exactly match the published ones, which are calculated at each sample point.

Appendix: indicators, weights and goalposts

Each indicator is listed once, with the share of its weight that counts in each domain and its effective weight in the index's mean level. Goalposts (Min, Ref, Max) are on the scale scored: the soft access score (0 to 1) where there is a threshold, the ladder where there is one, otherwise the value as measured. Ref is the study region average, and scores 0.

Indicator	Domains (share)	Scored as	Better	Min · Ref · Max	Effective weight
Natural blue space nearby	Ambient environment	ladder	higher	56.9 · 76.9 · 96.9	2.8%
Heat adaptive capability (GUHVI)	Ambient environment	as measured	higher	0 · 22 · 100	2.8%
Heat exposure (GUHVI)	Ambient environment	as measured	lower	0 · 48 · 100	2.8%
Heat sensitivity (GUHVI)	Ambient environment	as measured	lower	0 · 59 · 100	2.8%

Indicator	Domains (share)	Scored as	Better	Min · Ref · Max	Effective weight
<i>Daytime thermal comfort</i> <i>Included through walkability's thermal comfort attenuation. Choose walkability without attenuation to score it here.</i>	<i>Ambient environment</i>	<i>as measured</i>	<i>lower</i>	–	–
Parks within 300 m	Ambient environment (50%), Built environment (50%)	soft access, 300 m threshold	higher	-0.195 · 0.301 · 0.798	4.0%
Large public green space within 500 m	Ambient environment (50%), Built environment (50%)	as measured	higher	-0.419 · 0.081 · 0.581	4.0%
Vegetation cover (NDVI)	Ambient environment (50%), Built environment (50%)	as measured	higher	-0.251 · 0.082 · 0.416	4.0%
Proximity to major roads	Ambient environment (50%), Safety and protection (50%)	as measured	lower	-2,418 · 1,766 · 5,950	5.9%
Food within 1.5 km by low-stress cycling	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)	soft access, 1.5 km threshold	higher	0.059 · 0.556 · 1.053	6.2%
Block size (perimeter)	Built environment (33%), Mobility and transport (33%), Safety and protection (33%)	as measured	lower	-3,858 · 858 · 5,574	6.2%
Fire station within 3 km	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)	soft access, 3 km threshold	higher	0.099 · 0.596 · 1.092	6.2%
Police station within 2 km	Safety and protection (33%), Mobility and transport (33%), Built environment (33%)	soft access, 2 km threshold	higher	0.119 · 0.615 · 1.112	6.2%
Walkability (300 m), attenuated by thermal comfort	Built environment (33%), Mobility and transport (33%), Social infrastructure (33%)	as measured	higher	-0.022 · 0.315 · 0.651	3.7%
250 m or more from petrol stations	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	as measured	higher	0.421 · 0.921 · 1.421	3.6%
Convenience stores within 300 m	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	soft access, 300 m threshold	higher	0.184 · 0.680 · 1.177	3.6%
Fresh food diversity within 500 m	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	as measured	higher	-0.165 · 0.271 · 0.707	3.6%
Fresh food within 300 m	Social infrastructure (33%), Mobility and transport (33%), Economic development (33%)	soft access, 300 m threshold	higher	-0.089 · 0.408 · 0.905	3.6%
Job diversity	Economic development	as measured	higher	-0.236 · 0.264 · 0.764	5.0%
Manufacturing within 1 km	Economic development	as measured	higher	0.437 · 0.937 · 1.437	5.0%
Public transport within 300 m	Mobility and transport	soft access, 300 m threshold	higher	-0.432 · 0.064 · 0.561	4.2%
Community, culture and recreation diversity within 500 m	Social infrastructure	as measured	higher	-0.330 · 0.164 · 0.657	1.7%

Indicator	Domains (share)	Scored as	Better	Min · Ref · Max	Effective weight
Community, culture and recreation within 300 m	Social infrastructure	soft access, 300 m threshold	higher	-0.075 · 0.422 · 0.919	1.7%
Care service diversity within 500 m	Social infrastructure	as measured	higher	-0.439 · 0.061 · 0.561	1.7%
Care services within 300 m	Social infrastructure	soft access, 300 m threshold	higher	-0.335 · 0.162 · 0.658	1.7%
Education diversity within 500 m	Social infrastructure	as measured	higher	-0.230 · 0.270 · 0.770	1.7%
Health service diversity within 500 m	Social infrastructure	as measured	higher	-0.189 · 0.311 · 0.811	1.7%
Primary healthcare within 300 m	Social infrastructure	soft access, 300 m threshold	higher	-0.357 · 0.140 · 0.636	1.7%
Primary schools within 300 m	Social infrastructure	soft access, 300 m threshold	higher	-0.329 · 0.167 · 0.664	1.7%

Further reading

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